HACKNOVA 1.0 · 2026 · HBTU KANPUR · DEPARTMENT OF MATHEMATICS

# PortfolioIQ

Equity Risk & Returns Challenge — Prayas Capital

Team: Team 14 | Data: 490 trading days | Period: Jan 2023 – Dec 2024 | Universe: 15 NSE Stocks + Nifty 50 | Sectors: Banking · IT · Pharma

TASK 01

# Data Acquisition & Quality

**490**
Trading Days (2023–2024)

**16**
Tickers (15 Stocks + Nifty)

**0**
Missing Days (All Stocks)

**100%**
Data Quality Pass Rate

### BANKING SECTOR

| | |
|---|---|
| HDFCBANK | 490 days ✅ |
| ICICIBANK | 490 days ✅ |
| SBIN | 490 days ✅ |
| KOTAKBANK | 490 days ✅ |
| AXISBANK | 490 days ✅ |

### IT SECTOR

| | |
|---|---|
| TCS | 490 days ✅ |
| INFY | 490 days ✅ |
| WIPRO | 490 days ✅ |
| HCLTECH | 490 days ✅ |
| TECHM | 490 days ✅ |

### PHARMA SECTOR

| | |
|---|---|
| SUNPHARMA | 490 days ✅ |
| DRREDDY | 490 days ✅ |
| CIPLA | 490 days ✅ |
| DIVISLAB | 490 days ✅ |
| APOLLOHOSP | 490 days ✅ |

**Source:** yfinance Python library · Adjusted Close for all return calculations · NSE standard 252 trading days/year · Max consecutive gap verified: 0 days · All holidays treated as non-trading days per NSE calendar

TASKS 02 & 03

# Risk, Return & Technical Signals

| TICKER | ANN. RETURN | VOLATILITY | SHARPE | BETA | MAX DRAWDOWN | SMA SIGNAL |
|---|---|---|---|---|---|---|
| SUNPHARMA | +40.24% | 17.39% | 1.940 | 0.474 | -12.24% | Golden ✦ |
| HCLTECH | +43.26% | 21.38% | 1.719 | 0.789 | -21.66% | Golden ✦ |
| TECHM | +36.85% | 25.14% | 1.207 | 0.875 | -16.22% | Golden ✦ |
| DRREDDY | +29.36% | 19.46% | 1.175 | 0.430 | -15.61% | Death ✕ |
| DIVISLAB | +36.07% | 26.28% | 1.125 | 0.454 | -21.45% | Golden ✦ |
| APOLLOHOSP | +29.50% | 21.76% | 1.057 | 0.565 | -14.87% | Golden ✦ |
| ICICIBANK | +21.48% | 18.34% | 0.817 | 0.971 | -9.45% | Golden ✦ |
| WIPRO | +25.44% | 24.17% | 0.784 | 1.075 | -19.53% | Golden ✦ |
| HDFCBANK | +5.98% | 19.82% | -0.026 | 1.041 | -19.91% | Golden ✦ |
| KOTAKBANK | -2.31% | 20.41% | -0.432 | 0.865 | -23.15% | Death ✕ |

📌 MISPRICED STOCK

**HDFCBANK** — highest market-cap banking stock yet Sharpe of -0.026 vs ICICIBANK's 0.817. Substantially underperforms peers at similar volatility (~19-20%). It appears mispriced relative to it[illegible] market risk exposure.

⚠ MOST CORRELATED PAIR

**TCS & INFY** — correlation of 0.686. Both are large-cap IT exporters reacting identically to USD/INR moves [illegible] Holding both provides near-zero diversification benefit in the IT sector.

TASK 04

# Portfolio Construction

## Portfolio A — Equal Weight

₹66,667 × 15 stocks = ₹10,00,000

| | |
|---|---|
| Annualised Return | +25.10% |
| Annualised Volatility | 11.43% |
| Sharpe Ratio | 1.627 |
| Max Drawdown | -8.27% |
| Sector Exposure | 33% / 33% / 33% |

## Portfolio B — Recommended ★

Optimised allocation · 11 stocks · ₹10,00,000

| | |
|---|---|
| Annualised Return | +30.84% |
| Annualised Volatility | 11.18% |
| Sharpe Ratio | 2.178 |
| Max Drawdown | -7.80% |
| Sector Exposure | 21% / 29% / 50% |

### PORTFOLIO B — SECTOR ALLOCATION

| | |
|---|---|
| Banking | 21% |
| IT | 29% |
| Pharma | 50% |

### VS NIFTY 50 BENCHMARK

| | |
|---|---|
| Portfolio B Return | 30.84% |
| Portfolio A Return | 25.10% |
| Nifty 50 Return | 14.45% |

TASK 05

# Automation via n8n — Option B

**Daily Risk Digest Pipeline** — Runs every weekday at 9:00 AM IST. Fetches 3-month OHLCV for all Portfolio B stocks from Yahoo Finance. Computes 5-day rolling return, 1-day change, and SMA-50 crossover detection. Sends structured email digest to fund manager.

**Schedule** (9AM Mon–Fri) → **Set Tickers** (13 Portfolio B stocks) → **Split Loop** (1 per ticker) → **HTTP Request** (Yahoo Finance API) → **Code Node** (Compute metrics) → **Gmail** (Send digest)

## EMAIL DIGEST CONTAINS

- Top performer (5-day rolling return)
- Worst performer (5-day rolling return)
- Today's top gainer & top loser (1-day)
- NEW SMA-50 crossover alerts
- Full 13-stock scorecard table

## TECHNICAL SPECS

- Platform: n8n Cloud (app.n8n.cloud)
- Trigger: Cron 30 3 * * 1-5 (UTC)
- Data: Yahoo Finance v8 API (no key)
- Auth: Gmail OAuth2
- Submitted: JSON export + canvas screenshot

TASK 06

# ML Outperformance Classifier

### XGBoost — Model 1

| | |
|---|---|
| Accuracy | 46.69% |
| F1 Score | 44.07% |
| ROC-AUC | 0.4955 |
| CV F1 (5-fold) | 55.65% |

### Random Forest — Model 2 ★

| | |
|---|---|
| Accuracy | 52.28% |
| F1 Score | 44.85% |
| ROC-AUC | 0.5382 |
| CV F1 (5-fold) | 53.0% |

### Top 5 Feature Importances (RF)

### Per-Sector AUC (Random Forest)

| Banking | IT | Pharma |
|---|---|---|
| 0.421 | 0.692 | 0.515 |
| Below baseline | Strong signal | Slight edge |

**Interpretation:** IT sector AUC of 0.692 indicates that trend regime features (SMA crossover, beta) are highly predictive for IT stocks — likely due to USD/INR sensitivity creating persistent momentum. Banking sector near-random reflects RBI policy [illegible] *is pattern recognition on market features, not price prediction.*